

Air-Gapped Agentic AI Workbench

TECHNICAL SPECIFICATION & END-TO-END WORKFLOW GUIDE

Target Domain: Air-Gapped PSUs / Defence / Heavy Engineering

Network Policy: 100% Verified Zero-Egress (0.00 B/s)

Hardware Target: NVIDIA RTX 3060 (6GB VRAM) / 16GB System RAM

Runtime Stack: Local Stock Ollama + FastAPI + bwrap Sandboxes

1. Executive Summary & Core Mission

The **Air-Gapped Agentic AI Workbench** is a self-hosted autonomous intelligence system designed specifically for strict on-premise industrial, defence, and PSU environments. In these facilities, proprietary CAD blueprints, Piping & Instrumentation Diagrams (P&IDs;), sensor telemetry, and executive approval notes cannot leave the physical local network.

The workbench reconciles complex agentic workflows with consumer-grade hardware constraints (6GB VRAM on an NVIDIA RTX 3060) through a **high-precision prompt router**, a **deterministic multi-tool sandbox**, an **autonomous reflection loop**, and a **zero-egress hardware egress meter**.

Core System Guarantees:

- **Complete Data Sovereignty:** 100% on-device local execution using stock Ollama model serving.
- **Hardware Concurrency:** Dynamic memory swapping constrained to ≤ 2 active resident models.
- **Auditable Tooling:** Deterministic formulas in spreadsheets (=SUM()) and reproducible .docx/.pptx deliverables.
- **Kernel Sandbox:** Untrusted Python code executed in isolated Linux network namespaces (bwrap --unshare-net).

2. End-to-End System Architecture

Layer	Subsystem & Technology	Responsibilities & Isolation Boundary
User Interface Port 5173	Modern React / Vite Web UI TailwindCSS, Lucide Icons	Provides prompt interaction, live step timeline via SSE, visual deliverable downloads, real-time egress speedometer, and dynamic model hot-registration modal.
Control Plane Port 8000	FastAPI Orchestrator Async Python, Pydantic v2	Hosts Task Classifier, Model Selector, State Graph Executor, SQLite memory store, and SHA-256 immutable audit logger.
Tool Sandboxes Local OS	Kernel Isolated Workers bwrap, openpyxl, python-docx	Executes untrusted code in zero-network namespaces ('--unshare-net'), builds audit spreadsheets, and renders executive Word / PowerPoint briefs.
Inference Layer Port 11434	Stock Ollama Runtime 6GB VRAM Tuned ('MAX_LOADED=2')	Hosts quantized LLMs (Qwen 2.5 Coder 7B, Llama 3.1 8B, DeepSeek R1 7B, Moondream 1.8B, Nomic Embed) with automatic VRAM eviction.

3. Workflow 1: Prompt Ingestion, Classification & Model Routing

Model routing is managed by orchestrator/router/classifier.py and selector.py. The system executes a deterministic 5-stage decision ladder to assign the lowest-VRAM, highest-accuracy model for the incoming prompt without human intervention:

Stage 1: Manual Override Check	If the user explicitly selects a model via the UI dropdown, classification rules are bypassed and the request is routed directly to the selected model tag.
Stage 2: Document Context Rule	Uploading non-image documents (e.g., CSV, XLSX, PDF, DOCX, TXT) automatically routes to Llama 3.1 8B to leverage its 128,000 token context window .
Stage 3: Regex Pattern Matrix	Prompt text is evaluated against pre-compiled regex matrices for coding, vision, spreadsheet math, document drafting, and RAG retrieval.
Stage 4: Capability Tag Resolution	The detected task type is matched against capability tags declared in the dynamic model registry (registry.yaml).
Stage 5: VRAM Budget Optimization	If multiple models satisfy the capability, candidates are sorted ascending by vram_gb, selecting the most lightweight model to prevent GPU OOM.

Model Capability & Routing Registry Matrix

Task Category	Trigger Keywords & Regex Patterns	Target Model	Ollama Tag	VRAM
Vision / OCR	Image files (.png, .jpg), blueprints, P&ID; scans, drawings, OCR	vision-primary	moondream	2.0 GB
Coding & Sandbox	python, script, def, regex, execute, debug, unit test, algorithm	coding-primary	qwen2.5-coder:7b	4.7 GB
Math & Physics	stress tensor, fatigue limit, thermal formula, calculus, engineering derivation	math-engineering	deepseek-r1:7b	4.7 GB
Documents & SOPs	docx, pptx, approval note, SOP manual, summarize, policy clause	reasoning-primary	llama3.1:8b (128k)	4.9 GB
Vector Search	ChromaDB background embedding generation and similarity retrieval	embeddings	nomic-embed-text	0.3 GB

Hot Model Registration Workflow (Zero-Downtime)

New models (e.g., custom fine-tuned defence models) can be hot-registered via POST /api/models/register without stopping the orchestrator. The registry automatically updates registry.yaml, syncs with Ollama's local tag storage, and makes the model instantly selectable in the routing ladder.

4. Workflow 2: Autonomous Agent Execution Loop (Plan-Act-Observe-Reflect)

The cognitive engine in orchestrator/agent/graph.py operates as a multi-stage state machine featuring dynamic task decomposition, step-by-step verification, and automatic self-correction upon failures:

Phase	Engine Action	Detailed Mechanics & Recovery Behavior
1. PLAN	Task Decomposition	Deconstructs user goal into ordered sub-steps. Evaluates tool dependencies and determines parameter payloads.
2. ACT	Tool Dispatch	Dispatches structured tool calls into the secure sandbox (e.g. Python code execution, openpyxl spreadsheet generation, docx drafting).
3. OBSERVE	Output Capture	Captures stdout, stderr, process return codes, and filesystem artifacts into structured SQLite step records.
4. REFLECT	Validation & Reflection	Evaluates whether the step output satisfies acceptance criteria. Detects syntax errors, missing columns, or calculation anomalies.
5. CORRECT	Self-Correction Loop	If an error occurs, feeds the traceback back to the reasoning model, modifies the script, and retries up to 3 times before failing gracefully.
6. SYNTHESIZE	Deliverable Packaging	Produces executive summary text and publishes download links for generated .xlsx, .docx, or .pptx files.

5. Workflow 3: Secure Sandboxed Tool Execution Pipeline

Tool Name	Implementation Module	Security & Operational Mechanism
Code Sandbox code_execute	orchestrator/tools/sandbox.py	Executes untrusted Python via Linux kernel namespace isolation (bwrap --unshare-net --ro-bind / /) with timeout enforcement (15s default), 0 external sockets, and restricted deliverable outputs.
Spreadsheet Engine spreadsheet_create	orchestrator/tools/spreadsheet.py	Uses openpyxl to programmatically build styled Excel workbooks (.xlsx). Embeds native Excel formulas (e.g. =SUM(), =AVERAGE()) to guarantee mathematical auditability.
Document Generator docgen_approval_note docgen_presentation	orchestrator/tools/docgen.py	Generates production-grade Word documents (.docx) formatted with formal ministry/defence headers, reference numbers, risk tables, and PowerPoint decks (.pptx).
Filesystem Manager file_read / file_write	orchestrator/tools/files.py	Restricts file reads and writes strictly within the allowlisted workspace directory. Enforces path traversal defenses (blocks ../ and symlink escapes).

6. Workflow 4: Multimodal Ingestion & Air-Gapped RAG

The knowledge management pipeline in orchestrator/rag/ and orchestrator/ingestion/ processes proprietary industrial documents, scanned PDF reports, and P&ID; diagrams locally without cloud APIs:

- **1. Visual OCR Extraction:** Scanned diagrams and engineering blueprints are processed using **Moondream** (1.8B VLM) or local PyMuPDF extraction, converting visual data into structured textual tags.
- **2. Semantic Chunking:** Text documents are partitioned into overlapping chunks (512 tokens with 64-token stride) preserving technical paragraph integrity.
- **3. Local Embedding Generation:** Embeddings are computed on-device using nomic-embed-text (8192 context window, 0.3GB VRAM footprint).
- **4. ChromaDB Storage:** Dense vectors are stored in a persistent local ChromaDB instance at data/chroma_db/.
- **5. Hybrid Lexical & Semantic Retrieval:** Queries execute a reciprocal rank fusion of BM25 exact keyword matching and Cosine vector similarity, returning exact source citations (document name, page number, section header).

7. Workflow 5: Zero-Egress Air-Gap Security & Audit Trail

- **Real-Time Egress Monitoring:** The background service (network_monitor/) polls Linux socket tables and network interface statistics, reporting outbound bytes per second directly to the Web UI Egress Meter.
- **Zero-Egress Live Proof Command:** System administrators and audit judges can verify network silence at any moment with:
sudo tcpdump -i any not host 127.0.0.1 -n -c 20 (Yields 0 packets captured during agent runs).
- **Immutable Audit Logging:** Every user interaction, classified routing decision, prompt SHA-256 hash, model inference latency, and tool invocation is recorded in orchestrator/audit/logger.py.
- **Role-Based Access Control (RBAC):** JWT-based local authentication segregates permissions between standard Engineers, Technical Reviewers, and System Administrators.

Air-Gap Compliance Matrix:

- **External DNS Resolution:** Disabled / Blocked.
- **Telemetry & Analytics Callbacks:** 100% stripped from all client & server libraries.
- **Cryptographic Verification:** All local deliverables hashed with SHA-256 for chain-of-custody verification.

8. Workflow 6: Startup, Verification & Operational Commands

Operation / Workflow	Command / Endpoint	Description
Master Startup	<code>./start_workbench.sh</code>	Launches Ollama (6GB tuned), FastAPI backend (port 8000), and React Web UI (port 5173).
Automated Verification	<code>.venv/bin/python verify_all.py</code>	Runs full test suite validating all 6 Definition-of-Done criteria (Routing, Sandbox, Excel, DocGen, RAG, AirGap).
Model Hot Registration	<code>POST /api/models/register</code>	Dynamically registers new GGUF/Ollama models into registry.yaml without server downtime.
Agent Execution Stream	<code>POST /api/agent/run</code>	Executes multi-step agent loop with live Server-Sent Events (SSE) streaming of progress.
Network Egress Stream	<code>GET /api/network/egress</code>	Streams real-time network throughput and packet statistics to the dashboard speedometer.

9. Compliance Certification & Summary

Compliance Certification Summary: The Air-Gapped Agentic AI Workbench achieves 100% data sovereignty, full hardware compliance on single-GPU (6GB RTX 3060) workstations, and robust agentic task completion with zero external network connectivity. All workflows documented in this specification—including multi-model routing, isolated kernel sandboxing, deterministic spreadsheet generation, enterprise document drafting, multimodal OCR ingestion, and real-time egress monitoring—are fully implemented, verified, and operational within the local root index.
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